Why divide by 2π? Because it saves us from having to write 2π in lots of other places. Considering the importance of Planck's constant in quantum mechanics, it seems a little odd that it hasn't come up yet. We're going to correct that now.

In quantum mechanics, as in classical physics, the Hamiltonian is the mathematical object that represents the energy of a system. This raises a question that, if you are very alert, may have been a source of confusion. Take a good look at Eq. 4.9. It doesn't make dimensional sense. If you ignore |Ψ> on both sides of the equation, the units on the left side are inverse time. If the quantum Hamiltonian is really to be identified with energy, then the units on the right side are energy. Energy is measured in units of joules, or $kg \cdot m^2/s^2$. Evidently, I've been cheating a little bit. The resolution of this dilemma involves $\hbar$, a universal constant of nature, which happens to have units of $kg \cdot m^2/s$. A constant with these units is exactly what we need to make Eq. 4.9 consistent. Let's rewrite it with Planck's constant inserted in a way that makes it dimensionally consistent:

$$\hbar \frac{\partial |\Psi\rangle}{\partial t} = -i\mathbf{H}|\Psi\rangle. \qquad (4.10)$$

Why is it that $\hbar$ is such a ridiculously small number? The answer has much more to do with biology than with physics. The real question is not why $\hbar$ is so small; it's why you are so big. The units that we use reflect our own size. The origin of the meter seems to be that it was used to measure rope or cloth: it's about the distance from a person's nose to his or her outstretched fingers. A second is about as long as a heartbeat. And a kilogram is a nice weight to carry around. We use these units because they are convenient, but fundamental physics doesn't care that much about us. The size of an atom is about 10^{-10} meters. Why so small? That's the wrong question. The right one is: Why are there so many atoms in an arm? The reason is simply that to make a functioning, intelligent, unit-using creature, you need to put together a lot of atoms. Similarly, the kilogram is many times larger than an atomic mass because people don't carry around single atoms; they get lost too